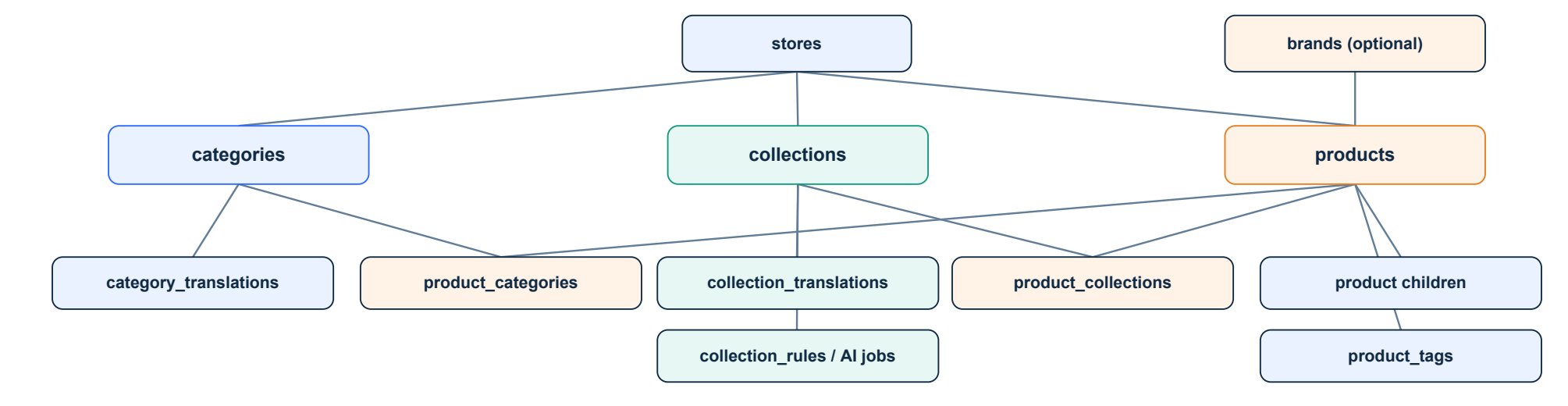


Category, Collection and Product Schema

Complete relation map and table structure generated from the existing ShopNxe PostgreSQL database. Snapshot: 2026-08-20.

26	5	4	PostgreSQL
Catalog tables	External anchor tables	Core schema migrations	Verified database



Every relation repeats store_id; composite foreign keys enforce Store isolation.

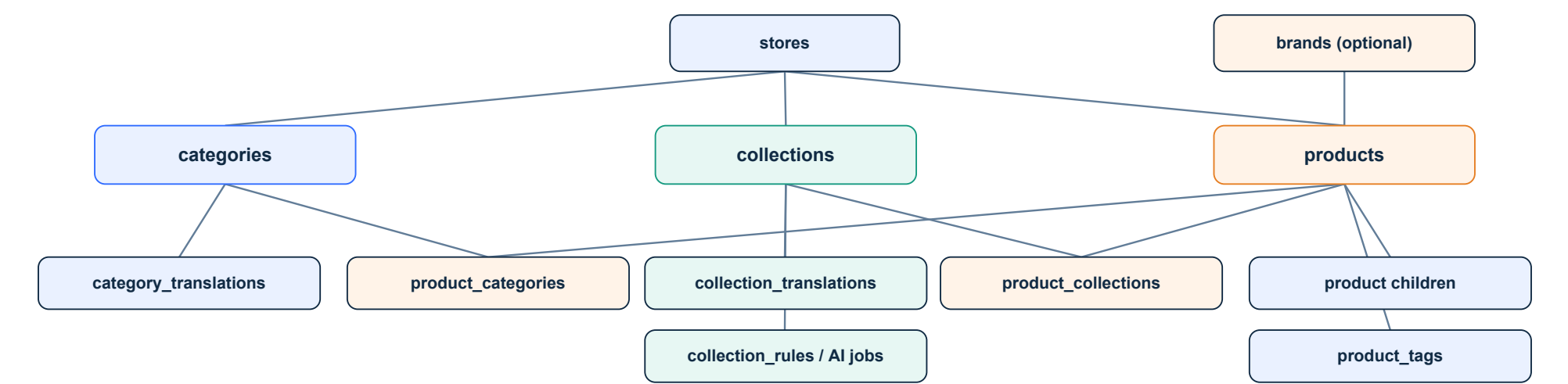
Solid paths indicate database-enforced foreign keys.

1. Conventions and tenant boundary

Store isolation: Every Catalog relationship carries store_id. Composite foreign keys such as (product_id, store_id) referencing products(id, store_id) prevent cross-Store assignments at the database level.

Notation	Meaning	Database behavior
PK	Primary key	Uniquely identifies a row.
UQ	Unique constraint or index	Prevents duplicate business identities.
FK	Foreign key	Enforces a valid parent row and defined delete behavior.
Nullable	Column may be NULL	Used for optional Brand, parent, Variant, image and lifecycle data.
lock_it	Translation overwrite lock	false by default; true preserves merchant-authored localized content.
Minor units	Integer money representation	Avoids floating-point price storage.

High-level Catalog relationships



Every relation repeats store_id; composite foreign keys enforce Store isolation.

Solid paths indicate database-enforced foreign keys.

2. Category aggregate

categories

Store-owned navigation taxonomy with an optional parent Category.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required FK to stores.id; Store deletion cascades.
parent_id	bigint	Nullable self-reference inside the same Store.
image_url	varchar(500)	Nullable shared Category image.
is_active	boolean	Required, default true.
sort_order	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, store_id). Self FK (parent_id, store_id) references categories(id, store_id). Deleting a parent sets only parent_id to NULL. Indexes: (store_id, is_active) and (store_id, parent_id, sort_order).

category_translations

Localized navigation, SEO, template and media presentation for a Category.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required FK to stores.id.
category_id	bigint	Required composite FK to categories.
locale	varchar(35)	Required locale key.
title	varchar(255)	Required localized title.
slug	varchar(255)	Required localized URL segment.
description	text	Nullable localized description.
seo_title	varchar(255)	Nullable SEO title.
seo_description	text	Nullable SEO description.
page_title	varchar(255)	Nullable storefront page title.
search_keywords	text	Nullable search keywords.
category_template	varchar(120)	Nullable storefront template identifier.
banner_url	varchar(500)	Nullable locale-specific banner.
image_url	varchar(500)	Nullable locale-specific image.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (category_id, locale). Unique (store_id, locale, slug). Composite Category FK cascades on delete.

product_categories

Many-to-many assignment between Products and Categories.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
category_id	bigint	Required composite FK to categories.
product_id	bigint	Required composite FK to products.
sort_order	integer	Required, default 0.
is_primary	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (category_id, product_id). Partial unique index (store_id, product_id) WHERE is_primary allows only one primary Category per Product. Parent deletion removes only the assignment.

3. Collection aggregate

collections

Store-owned merchandising groups populated manually, by rules or from AI-generated rules.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required FK to stores.id.
parent_id	bigint	Nullable same-Store parent Collection.
image_url	varchar(500)	Nullable Collection image.
is_active	boolean	Required, default true.
sort_order	integer	Required, default 0.
collection_type	varchar(20)	Default manual; manual, rule_based or ai_generated.
rules_match_type	varchar(10)	Default all; all or any.
ai_prompt	text	Nullable latest AI instruction.
ai_model	varchar(100)	Nullable latest model identifier.
ai_status	varchar(20)	Nullable; pending, processing, completed or failed.
ai_last_run_at	timestamptz	Nullable latest run time.
ai_error_message	text	Nullable latest operator-facing error.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, store_id). Self FK sets parent_id to NULL when a parent is deleted. Indexes: (store_id, collection_type) and (store_id, parent_id, sort_order).

collection_translations

Localized Collection content and SEO metadata.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
collection_id	bigint	Required composite FK to collections.
locale	varchar(35)	Required locale key.
title	varchar(255)	Required localized title.
slug	varchar(255)	Required localized URL segment.
description	text	Nullable localized description.
seo_title	varchar(255)	Nullable SEO title.
seo_description	text	Nullable SEO description.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (collection_id, locale). Unique (store_id, locale, slug). Collection deletion cascades.

collection_rules

Ordered structured conditions that define automatic membership.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
collection_id	bigint	Required composite FK to collections.
field	varchar(50)	Required Product field identifier.
operator	varchar(20)	Required comparison operator.
value	varchar(255)	Required serialized operand.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, store_id). Index (store_id, collection_id, position). Collection deletion cascades.

3. Collection aggregate - automation and membership

collection_ai_jobs

Append-style audit history for AI Collection generation.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
collection_id	bigint	Required composite FK to collections.
prompt	text	Required prompt used for the run.
model	varchar(100)	Required model identifier.
status	varchar(20)	Default pending; pending, processing, completed or failed.
result_rules	jsonb	Nullable raw structured output.
matched_count	integer	Nullable matched Product count.
error_message	text	Nullable failure detail.
tokens_used	integer	Nullable usage accounting.
completed_at	timestamptz	Nullable terminal time.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

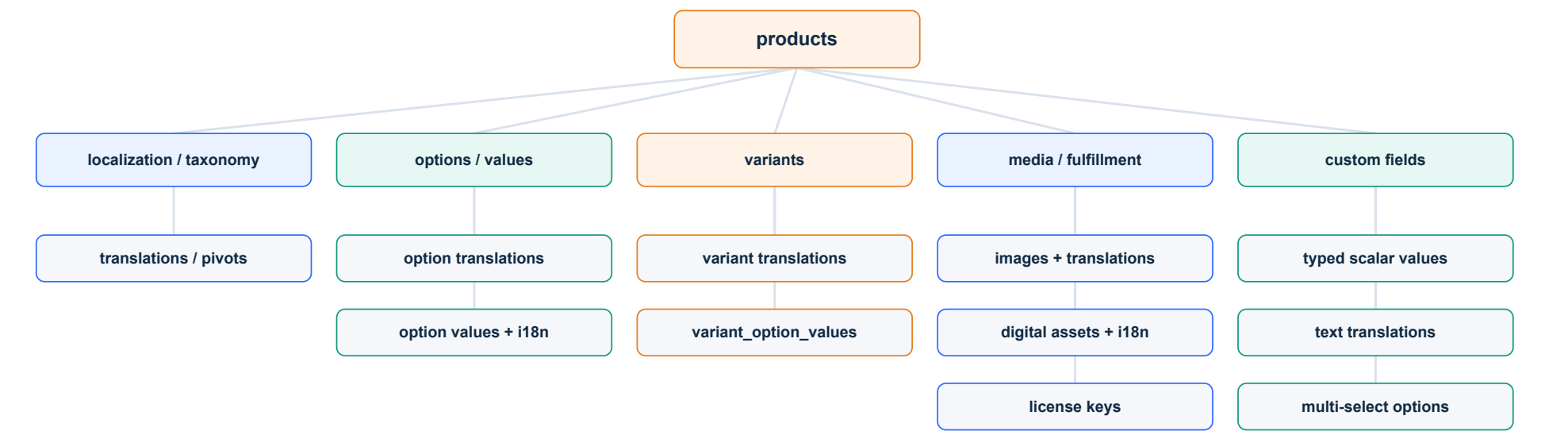
Keys and relations: Unique (id, store_id). Index (store_id, collection_id, created_at). The row records a run; application code must normalize rules and Product assignments.

product_collections

Many-to-many Product membership with automation metadata. PK (collection_id, product_id); indexes support Product lookup and Collection sorting. Parent deletion removes only the assignment. Regeneration should replace unpinned automated rows while preserving pinned merchant decisions.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
collection_id	bigint	Required composite FK to collections.
product_id	bigint	Required composite FK to products.
sort_order	integer	Required, default 0.
added_by	varchar(10)	Default manual; manual, rule or ai.
is_pinned	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

4. Product aggregate



Direct Product ownership cascades on delete; nullable Variant links detach where noted.

products

The Product root contains Store-wide, non-localized behavior and publication state.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required FK to stores.id.
brand_id	bigint	Nullable same-Store Brand reference.
vendor	varchar(255)	Nullable supplier text.
product_type	varchar(255)	Nullable merchant Product type.
fulfillment_type	varchar(20)	Default physical; physical, digital, software or service.
track_inventory	boolean	Required, default true.
status	varchar(20)	Default draft; draft, active or archived.
has_variants	boolean	Required, default false.
published_at	timestamptz	Nullable publication time.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, store_id). Brand deletion sets brand_id to NULL. Indexes: (store_id, status), (store_id, fulfillment_type), and (store_id, brand_id).

4. Product localization and taxonomy

product_translations

Localized storefront content and SEO metadata.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
locale	varchar(35)	Required locale key.
title	varchar(255)	Required localized title.
slug	varchar(255)	Required localized URL segment.
description	text	Nullable localized description.
seo_title	varchar(255)	Nullable SEO title.
seo_description	text	Nullable SEO description.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (product_id, locale). Unique (store_id, locale, slug). Product deletion cascades.

product_tags

Many-to-many Product and Tag assignment.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
tag_id	bigint	Required composite FK to tags.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (product_id, tag_id). Index (store_id, tag_id). Deleting either parent removes the assignment.

Pivot	Primary key	Additional state	Delete behavior
product_categories	(category_id, product_id)	sort_order, is_primary	Cascade assignment
product_collections	(collection_id, product_id)	sort_order, added_by, is_pinned	Cascade assignment
product_tags	(product_id, tag_id)	Timestamps	Cascade assignment

5. Product options and option values

Table	Key	Important columns	Purpose
product_options	id PK; Product FK	position=0	Defines Color, Size or another dimension.
product_option_translations	PK (option_id, locale)	name, lock_it=false	Localized option name.
product_option_values	id PK; composite Option FK	product_id, option_id, position=0	Defines Red, Blue, Small or similar choices.
product_option_value_translations	PK (option_value_id, locale)	value, lock_it=false	Localized choice label.

product_options

Product-scoped option definitions.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, store_id) and (id, product_id, store_id). Index (store_id, product_id, position).

product_option_values

Ordered values that belong to a Product option.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required Product scope.
option_id	bigint	Required composite FK to product_options.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Composite FK (option_id, product_id, store_id) ensures the value belongs to the same Product. Unique (id, product_id, store_id).

6. Product variants and selections

product_variants

Purchasable Product combinations with price, inventory and fulfillment dimensions.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
sku	varchar(100)	Nullable; unique per Store when present.
barcode	varchar(100)	Nullable.
price_amount_minor	bigint	Required, non-negative.
compare_at_price_amount_minor	bigint	Nullable, non-negative.
msrp_amount_minor	bigint	Nullable, non-negative.
cost_per_item_amount_minor	bigint	Nullable, non-negative.
currency_code	character(3)	Required, three uppercase letters.
inventory_qty	integer	Required, default 0.
inventory_policy	varchar(20)	Default deny; deny or continue.
weight_grams	integer	Nullable.
height	numeric(12,4)	Nullable.
width	numeric(12,4)	Nullable.
depth	numeric(12,4)	Nullable.
dimension_unit	varchar(10)	Required, default cm.
requires_shipping	boolean	Required, default true.
taxable	boolean	Required, default true.
call_for_price	boolean	Required, default false.
image_id	bigint	Nullable representative Product image.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, product_id, store_id). Index (store_id, product_id, position). Deleting the Product cascades; deleting a representative image sets image_id to NULL.

6. Variant translations and selected values

product_variant_translations

Localized optional Variant titles.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
variant_id	bigint	Required composite FK to product_variants.
locale	varchar(35)	Required locale key.
title	varchar(255)	Nullable localized title.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (variant_id, locale). Variant deletion cascades.

variant_option_values

Join table that records each option value selected by a Variant.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
product_id	bigint	Required Product scope.
variant_id	bigint	Required composite FK to product_variants.
option_value_id	bigint	Required composite FK to product_option_values.
created_at	timestamptz	Required, default CURRENT_TIMESTAMP.

Keys and relations: PK (variant_id, option_value_id). Both composite FKs include product_id and store_id, preventing a Variant from selecting a value from another Product or Store.

Money storage: Variant prices use integer minor units. For a two-decimal currency, 250000 represents 2,500.00. This avoids floating-point rounding in the database.

7. Product images

product_images

Ordered Product images with optional Variant specialization.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
variant_id	bigint	Nullable same-Product Variant reference.
url	varchar(500)	Required image locator.
width	integer	Nullable pixel width.
height	integer	Nullable pixel height.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (id, product_id, store_id). Indexes by Product position and Variant. Deleting the Product cascades; deleting the Variant sets variant_id to NULL.

product_image_translations

Localized alternative text for an image.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
image_id	bigint	Required composite FK to product_images.
locale	varchar(35)	Required locale key.
alt_text	varchar(255)	Nullable accessibility text.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (image_id, locale). Image deletion cascades.

8. Digital assets and licenses

product_digital_assets

Downloadable files attached to a Product or optional Variant.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
variant_id	bigint	Nullable same-Product Variant reference.
file_url	varchar(500)	Required file locator.
file_name	varchar(255)	Required file name.
file_size_bytes	bigint	Nullable file size.
file_type	varchar(50)	Nullable file type.
download_limit	integer	Nullable allowed download count.
link_expires_after_days	integer	Nullable link lifetime.
position	integer	Required, default 0.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Indexes by Product position and Variant. Product deletion cascades; Variant deletion sets variant_id to NULL.

product_digital_asset_translations

Localized display name and description for downloadable files.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
digital_asset_id	bigint	Required composite FK to product_digital_assets.
locale	varchar(35)	Required locale key.
display_name	varchar(255)	Nullable localized display name.
description	text	Nullable localized description.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (digital_asset_id, locale). Digital asset deletion cascades.

8. Product license keys

product_license_keys

Store-scoped license inventory for software and license-based Products.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
variant_id	bigint	Nullable same-Product Variant reference.
key_code	varchar(255)	Required; unique per Store.
status	varchar(20)	Default available; available, assigned, revoked or expired.
max_activations	integer	Required, default 1; must be greater than zero.
assigned_to_order_id	bigint	Nullable and indexed; no Order FK currently exists.
assigned_at	timestamptz	Nullable assignment time.
expires_at	timestamptz	Nullable expiry time.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique (store_id, key_code). Indexes by Product/status and Variant/status. Product deletion cascades; Variant deletion sets variant_id to NULL.

Order integration gap: assigned_to_order_id is an indexed bigint but is not protected by a database foreign key. Order existence and Store consistency must currently be enforced by application code.

9. Product custom-field values

product_custom_field_values

Typed values attached at Product level or optionally overridden for a Variant.

Column	PostgreSQL type	Null, default, key or meaning
id	bigint	Auto-increment PK.
public_id	character(26)	Required, unique public ULID.
store_id	bigint	Required Store boundary.
product_id	bigint	Required composite FK to products.
variant_id	bigint	Nullable same-Product Variant override.
definition_id	bigint	Required composite FK to custom_field_definitions.
value_number	numeric(18,4)	Nullable numeric scalar.
value_boolean	boolean	Nullable boolean scalar.
value_date	date	Nullable date scalar.
value_option_id	bigint	Nullable single-select option; deletion restricted while used.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: Unique expression scope (store_id, definition_id, product_id, COALESCE(variant_id, 0)). A CHECK permits at most one scalar column. Product, Variant or Definition deletion cascades as applicable.

product_custom_field_value_translations

Localized text value for translatable custom fields.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
value_id	bigint	Required composite FK to product_custom_field_values.
locale	varchar(35)	Required locale key.
value_text	text	Required localized text.
lock_it	boolean	Required, default false.
created_at	timestamptz	Nullable audit timestamp.
updated_at	timestamptz	Nullable audit timestamp.

Keys and relations: PK (value_id, locale). Parent custom-field value deletion cascades.

product_custom_field_value_options

Multi-select join between a Product custom-field value and allowed options.

Column	PostgreSQL type	Null, default, key or meaning
store_id	bigint	Required Store boundary.
definition_id	bigint	Required definition scope.
value_id	bigint	Required composite FK to product_custom_field_values.
option_id	bigint	Required composite FK to custom_field_options.
created_at	timestampz	Required, default CURRENT_TIMESTAMP.

Keys and relations: PK (value_id, option_id). Composite definition and Store keys prevent selecting an option from another field or Store.

10. Foreign-key delete behavior

Deleted parent	Affected relation	Database result
Store	All Category, Collection and Product tables	Cascade delete.
Parent Category	categories.parent_id	Set parent_id to NULL; keep child.
Parent Collection	collections.parent_id	Set parent_id to NULL; keep child.
Brand	products.brand_id	Set brand_id to NULL; keep Product.
Category	category translations and product_categories	Cascade children and assignments.
Collection	translations, rules, AI jobs and product_collections	Cascade children and assignments.
Product	All Product-owned rows	Cascade through translations, pivots, options, variants, media, licenses and custom values.
Variant	Variant translations and selections	Cascade delete.
Variant	image, digital asset and license optional links	Set variant_id to NULL.
Variant	Variant-specific custom-field values	Cascade delete.
Product image	product_variants.image_id	Set image_id to NULL.
Custom-field definition	Product custom-field values	Cascade delete.
Single-select option in use	value_option_id	Restrict deletion.
Multi-select option	product_custom_field_value_options	Cascade pivot row.

Deletion principle: Core entities delete their owned dependent rows. Shared entities such as Brands, Categories, Collections and Tags either detach or remove only pivots so unrelated Catalog records survive.

11. Constraints and index summary

Area	Constraint or index	Purpose
Category	(store_id, locale, slug) unique	Localized Category URLs are unique per Store.
Category	One primary Category partial index	Only one product_categories row can be primary per Product.
Collection	Type and status CHECK constraints	Restricts Collection and AI lifecycle values.
Collection	Store/type and Store/parent/sort indexes	Supports listing and hierarchy ordering.
Product	Status, fulfillment and Brand indexes	Supports Store-scoped management filters.
Product	(store_id, locale, slug) unique	Localized Product URLs are unique per Store.
Variant	Partial unique (store_id, sku)	A non-null SKU cannot repeat inside a Store.
Variant	Currency and non-negative price CHECKs	Protects monetary integrity.
Variant	Product/position index	Supports ordered Product Variant retrieval.
Media	Product/position and Variant indexes	Supports ordered galleries and Variant lookups.
License	Store/key unique and status indexes	Protects key uniqueness and assignment queries.
Custom fields	Expression unique Product/Variant scope	One field value per Product or Variant scope.
Custom fields	Scalar num_nonnulls CHECK	Prevents multiple scalar representations in one row.

External anchor tables

Anchor	Referenced by	Role
stores	Every Catalog table	Tenant ownership and cascade root.
brands	products.brand_id	Optional Product Brand.
tags	product_tags.tag_id	Shared Product classification.
custom_field_definitions	product_custom_field_values.definition_id	Defines field key and type.
custom_field_options	Single-select and multi-select values	Defines allowed choices.

12. Source and implementation notes

The structure in this document was checked against the live PostgreSQL database with Laravel's db:table inspection command. The following applied migrations define the schema:

#	Repository source
1	Modules/Catalog/database/migrations/2026_08_05_000100_create_catalog_taxonomy_tables.php
2	Modules/Catalog/database/migrations/2026_08_05_000200_create_catalog_product_tables.php
3	Modules/Catalog/database/migrations/2026_08_05_000300_create_catalog_variant_fulfillment_tables.php
4	Modules/Catalog/database/migrations/2026_08_05_000400_create_catalog_custom_field_value_tables.php
5	Modules/Catalog/database/migrations/2026_08_09_000100_add_lock_it_to_catalog_translation_tables.php
6	Modules/Catalog/database/migrations/2026_08_08_000100_add_page_title_and_search_keywords_to_category_translations_table.php
7	Modules/Catalog/database/migrations/2026_08_08_000200_add_category_template_to_category_translations_table.php
8	Modules/Catalog/database/migrations/2026_08_08_000300_add_banner_url_to_category_translations_table.php
9	Modules/Catalog/database/migrations/2026_08_17_000100_add_image_url_to_category_translations_table.php

Application coverage: The database schema is broader than current ORM and API coverage. Product currently exposes Store, Brand, translations and Categories in its model; Collection and several Product child aggregates still require application-layer models, services and endpoints before they can be fully managed.

Generated as a technical database reference. This document contains schema metadata only and does not include credentials or row data.